

Student 2 - Class Design & Java Concepts

Time: 3 minutes

"Thank you. Now I will explain how our project is designed and what Java concepts we used."

| Action: Move to center. Show class diagram slide.

"Our project has 6 classes. Each class has one job. This is called Separation of Concerns."

| Action: Point to class diagram. Count 6 on fingers.

"Item.java is our data model. It has 8 fields - id, type, name, category, date, location, contact, and status."

| Action: Point to Item class on diagram.

"StorageManager handles all file reading and writing. It reads from and writes to data/items.txt."

| Action: Point to StorageManager class. Show file path.

"ItemService is the brain. It handles posting, searching, and claiming items."

| Action: Point to ItemService class.

"Main.java is the user interface. It shows menus and reads your input."

| Action: Point to Main class.

"Now let me explain the Java concepts. First - Classes and Objects. A class is a blueprint. An object is the actual thing made from it."

| Action: Use hands to show blueprint vs object.

"Second - Encapsulation. We use getters to read data and only one setter for status. This protects our data."

| Action: Show getter/setter on diagram.

"Third - Java Streams. Instead of writing long loops, we use streams to filter items. For example, to get only lost items, we filter by type equals LOST."

| Action: Show stream code snippet on screen.

"Fourth - Lambda Expressions. These are short anonymous functions used with streams. Like i - arrow - i.getType equals LOST."

| Action: Point to lambda syntax on screen.

"Fifth - Try With Resources. This automatically closes files after reading or writing. It prevents resource leaks."

| Action: Show try-with-resources code.

"The class dependency is simple. Main uses ItemService. ItemService uses StorageManager. StorageManager uses Item. Clean and simple."

| Action: Trace the dependency flow on diagram.

"Now I will hand over to explain the actual implementation and code."

| Action: Gesture towards next presenter. Step back.
